

# HOSE

## Job Performance Requirements

4.3.10\*

4.5.2

5.3.1\*

**Note: All objectives in this section are testable.**

## Introduction

Subject: Hose

Note: ALL lengths, measurements and percentages must be verbalized.

1. All evolutions which involve making connections and couplings, require detection of a gasket being present and the swivel being functional. This shall be done prior to deployment and after use.
2. "Gasket, Swivel" must be verbalized.
3. Visually detect by looking down into the swivel section of the female coupling.
4. Physically detect by placing a finger inside the swivel section of the female coupling to remove. Bend and pinch gasket while observing for any discrepancies.
5. When inserting or replacing a gasket, ensure it is seated properly in the groove of the female couplings by pressing the edges of the gasket down.
6. Hose and Appliances shall NOT be used without a gasket present.
7. When loading hose in the bed of the engine, connect couplings "hand tight."
8. If couplings are leaking water excessively, utilize spanner wrenches to tighten couplings.
9. When loading large diameter supply hose, all couplings shall be laid allowing them to deploy without flipping over. To achieve this, make a "Dutchman" (a reverse bend or short fold in the hose, ensuring the coupling is laid toward the front of the hose bed).
10. When loading attack hose lines, the total length of hose shall be one hundred fifty (150) feet for the state practical exams.
11. All hose loads should be loaded uniformly and evenly with the edge of the bed.
12. Adjustments for hose loads may be necessary, due to the location of discharge outlets and the configuration of hose beds, including cross lays.
13. The BFST understands not all buildings are set up to perform all skills in the same manner. **Prior to the practical examination, the Assistant Superintendent of the Bureau of Fire Standards and Training is requesting that the Training Center Instructors inform the Field Representatives of any alternate methods of instruction so that the students will not lose points during testing.**
14. **When replacing a burst section of hose, have someone open the bale of the charged hose line to allow water to flow prior to skill/time starting. This is NOT part of the skill as a true burst section would have water flowing.**

## **Performance Objective**

**Subject:** Hose

**Task:** Coupling Hose; Foot Tilt Method

**Procedure:**

1. With the hose lying flat on the ground, place the arch of your foot directly behind and touching the male coupling.

**Note:** Folding the male coupling over and lying on top of the hose is accepted.

2. Push your foot down pinning the male coupling against the ground. This will secure, stabilize, and cause the coupling to face upward.
3. Grasp the female coupling, detect and verbalize "Gasket, Swivel."
4. Position the female against the male coupling.
5. Align the Higbee cuts utilizing the Higbee indicators located on the male shank and the female swivel.
6. If the Higbee indicators are not visible, place the female coupling against the male coupling and turn counterclockwise until the Higbee cuts mate.
7. Turn the swivel clockwise to demonstrate the connection.

## Performance Objective

Subject: Hose

Task: Uncoupling Hose; Knee Press Method

Procedure:

1. Position the connected coupling perpendicular to the ground, **with the male on the bottom and the female on top.**
2. Place your knee directly behind the female and press downward using your body weight. This relieves pressure exerted on the connected coupling.
3. Quickly snap the swivel in a counterclockwise direction, breaking and loosening the seal of the connected coupling.
4. **Remove your knee and body weight from the female.**
5. Utilize a foot tilt or over the hip technique to continue and demonstrate the uncoupling process.
6. Detect and verbalize "Gasket, Swivel."

## **Performance Objective**

Subject: Hose

Task: Flat Load

Procedure:

1. Arrange fire hose in which the hose lies flat with successive layers (tiers) one upon the other.
2. Lay the 1<sup>st</sup> length of hose flat against the side of the bed, with the coupling 12-18 inches from the front of the bed for LDH. This space is reserved for couplings.

Note: In a split hose bed, start the load against the partition (baffle) with the coupling hanging below the bed. This allows the coupling to connect with the last coupling of the load on the opposite side of the partition.

3. Fold the hose back laying on itself and bring it toward the front of the bed.
4. Fold the hose at an angle and bring the hose back to the rear of the bed. Ensure it is laid next to the first section loaded.
5. Continue loading the hose in this manner to the other side of the bed, creating the 1<sup>st</sup> tier of the load.
6. Continue to load the hose in succession creating multiple tiers.

Note: Every other tier should be loaded approximately two (2) inches shorter than the previous tier. This helps the load maintain a flat appearance throughout the entire process. It also presents a visual aid in verbalizing the different tiers to unload the hose correctly.

## **Performance Objective**

**Subject:** Hose

**Task:** Accordion Load

**Procedure:**

1. Arrange fire hose in which the hose lies on edge with successive folds adjacent to each other.

**Note:** The advantage of this load is firefighters can easily grasp several folds and place them on one (1) shoulder to carry.

2. Lay the 1<sup>st</sup> length of hose on its edge and against the side of the bed, with the coupling at the rear of the bed.

**Note:** In a split hose bed, start the load against the partition (baffle) with the coupling hanging below the bed. This allows the coupling to connect with the last coupling of the load on the opposite side of the partition.

3. Create a bend at the front of the bed and bring the hose back to the rear lying against the 1<sup>st</sup> length.
4. Fold the hose so the bend is even with the rear edge of the bed.
5. Lay the hose back to the front of the bed and continue loading in this manner to the other side of the bed.

**Note:** Stagger every other fold approximately two (2) inches from the previous fold. This helps the load maintain an accordion appearance throughout the entire process.

6. When the 1<sup>st</sup> tier is demonstrated, angle the hose up from last bend at the front of the bed toward the rear to begin the 2<sup>nd</sup> tier.
7. Make the 1<sup>st</sup> fold of the 2<sup>nd</sup> tier directly over the last fold of the 1<sup>st</sup> tier at the rear of the bed.
8. Continue with the 2<sup>nd</sup> and subsequent tiers in the same manner.

## Performance Objective

Subject: Hose

Task: Hydrant Connection

Procedure:

1. While working with the hydrant, do not stand in front of any outlet cap or lean over the top of hydrant.
2. Using a hydrant wrench, ensure caps not being utilized are tight before removing the cap to be used (**Tighten, Tighten, Loosen**). Remove the necessary cap.
3. Detect the hydrant for damage or debris.
4. Open the hydrant by turning the operating nut counterclockwise.
5. Flush hydrant verifying adequate water source, ensuring outlet port has full orifice of water flow and water is clear.
6. Close (shut down) the hydrant by turning the operating nut clockwise.
7. Detect the hose verifying the presence of a gasket and the swivel is functional.

Note: If additional hose is needed other than the supply line, a hydrant valve can be attached (to one of the other ports) to utilize lines without closing the hydrant.

8. Attach the hose to hydrant and await signal to charge the hose. **DO NOT open the hydrant and charge the hose until signaled by the apparatus operator.**
9. At the apparatus, attach the hose to the intake verifying "**Gasket, Swivel.**"
10. After opening the hydrant fully, proceed to the apparatus bringing the hydrant wrench with you. **DO NOT leave the hydrant wrench at the hydrant.**
11. While returning to the apparatus, remove any kinks in the hose line and tighten any couplings leaking water.

## Performance Objective

Subject: Hose

Task: Apply a Hose Clamp

Note: Hose clamps are utilized to stop the water flow without shutting down the water source for the following applications: To prevent charging the hose remaining in the bed during a forward lay; Replacing a burst section of hose; Extending a hose line; To prevent discharge of water prior to making connections during a reverse lay.

When possible, always try to shut off water at the source. It is a much safer action than to apply a hose clamp for water control.

Procedure:

1. Apply the clamp at least twenty (20) feet behind the apparatus.
2. Apply the clamp no closer than five (5) feet from the coupling and on the supply side.
3. Place the hose centered in the jaws to avoid pinching the hose.
4. Using a “press down” clamp, **position yourself on the supply side of the clamp.**
5. Using a “screw” clamp, **position yourself on the latched side (opposite the hinged side) and on the supply side of the clamp.**

## **Performance Objective**

**Subject:** Hose

**Task:** Forward Hose Lay

**Note:** This objective is performed by two (2) individuals (Hydrant person and Apparatus operator) simultaneously.

### **Procedure:**

1. Traveling in the direction of the incident, the apparatus should stop approximately ten (10) feet past the hydrant.

Hydrant person:

2. Take tools and appliances needed to demonstrate hydrant connection.
3. At the rear of the hose bed, grasp the female or sexless coupling and enough hose (approximately 30 feet) to reach the hydrant.
4. Pull the hose from the bed, then pass the hydrant on the street side and wrap the hydrant in a manner to restrain the hose when the apparatus proceeds to the incident.
5. While standing on the correct side of the hydrant, and once done opening and flushing the hydrant, signal the driver to proceed to the incident and close the hydrant.
6. When sufficient hose is laid on the ground, demonstrate the hydrant connection as previously instructed.

Apparatus operator:

7. When apparatus has arrived at the incident, apply a hose clamp behind the apparatus as previously instructed.
8. Signal the hydrant person to charge the hose line.
9. Remove enough hose from the bed to reach the intake of the apparatus.
10. Break the coupling and place the unused end back in the hose bed.
11. Attach the male or sexless coupling to the intake of the apparatus and remove the hose clamp.

## **Performance Objective**

**Subject:** Hose

**Task:** Reverse Hose Lay

**Note:** This objective is performed by two (2) or more personnel (Incident scene personnel and Apparatus operator) simultaneously.

**Procedure:**

1. Traveling in the direction of the water source, the apparatus will stop at the scene of the incident.

Incident scene personnel:

2. All equipment needed for the incident shall be removed and staged. Ensure enough hose is removed to accomplish necessary tasks.
3. Safely anchor the supply hose line, by kneeling or standing on the hose.
4. Signal the apparatus operator to proceed to the water source.
5. On scene, apply a hose clamp as previously instructed.
6. Signal the apparatus operator to charge the hose line.

Apparatus operator:

7. When the apparatus has arrived at the water source, demonstrate the hydrant connection as previously instructed.
8. Make all necessary hose and appliance connections for the incident.

## **Performance Objective**

**Subject:** Hose

**Task:** Pulling/Advancing Hose Lines from a Flat Load with 2<sup>1</sup>/<sub>2</sub> or 3-inch Hose;  
Shoulder Load

**Procedure:**

1. Stand on the tailboard or ground to reach the hose load.
2. Facing the hose bed, grasp the nozzle or coupling, with several folds using both hands. Pull the folds out of the hose bed, approximately four (4) feet.
3. While still grasping the load, turn to face away from the hose bed. Twist the load on your shoulder so the nozzle or coupling will be positioned against your body, and in front of you at the waist.
4. Walk away from the apparatus until enough hose has been removed on your shoulder to demonstrate the operation.

## **Performance Objective**

Subject: Hose

Task: Pre-connect attack line; Flat Load

Procedure:

1. Connect three (3) sections of hose together and connect the female coupling to the discharge. Lay the 1<sup>st</sup> length of hose flat in the hose bed.
2. Make a fold in the hose even with the edge of the bed and continue loading the hose with each tier on top of the previous fold.

Note: This can be loaded double stacked (side by side), to accommodate the configuration of the hose bed. Angle the hose and make a fold laying adjacent to the 1<sup>st</sup> fold, ensuring all tiers of folds are evenly stacked. Continue loading the hose in the same manner.

3. At approximately one-third ( $\frac{1}{3}$ ) the total length of the hose load, make a fold which extends beyond the hose bed, to serve as a handle to assist deploying the hose (See PO for deploying a Flat Load).
4. Continue loading the hose in the same manner, building each tier with folds stacked evenly. At approximately two-thirds ( $\frac{2}{3}$ ) the total length of the hose load, make another fold which extends beyond the hose bed, to serve as a second larger handle to assist deploying the hose (See PO for deploying a Flat Load).
5. Upon completion, connect the nozzle and lay it on top of the hose load at the rear of the bed.

**Note: PO's regarding hose loads are for instructional purposes. All hose loads have variations for loading and deploying. The BFST recognizes hose loads and deployment can be accomplished by using different methods and does not intend to evaluate these procedures.**

## **Performance Objective**

Subject: Hose

Task: Deploy a Pre-connect Flat Load

Procedure:

1. Slide one arm through the larger hose handle and grasp the smaller hose handle with the same hand.

Note: This allows hose to NOT be dropped next to the engine and cause a delay or time constraint for candidates during testing.

2. Grasp the nozzle with the opposite hand.
3. Pull the entire hose load demonstrates from the hose bed.
4. Advance hose toward the fire or entry point of structure.
5. As hose becomes taut, release the smaller handle and continue to deploy.
6. Again, as hose becomes taut, release the larger handle and continue to deploy.
7. When hose is fully deployed, signal the driver operator for water and be prepared for movement in the hose line due to charging and pressure.
8. Bleed hose line of any air before continuing and be prepared for a backward force (nozzle reaction) when opening the bale and flowing water.

## **Performance Objective**

Subject: Hose

Task: Pre-connect attack line; Minuteman Load

Procedure:

1. Connect the female coupling of one (1) section of hose to the discharge. Lay the hose flat in the bed, and place remaining section of hose at the front and out of the bed.
2. Connect the other two (2) sections of hose together with the nozzle.
3. Place the nozzle on top of the 1<sup>st</sup> hose, at the rear of the bed.

Note: This can be loaded double stacked (side by side), to accommodate the configuration of the hose bed. Angle the hose and make a fold laying adjacent to the nozzle, ensuring all tiers of folds are evenly stacked. Continue loading the hose in the same manner.

4. Load the hose to the front, then back to the rear of the bed, making folds evenly with the nozzle. Continue loading the hose in the same manner until the female coupling is reached.
5. Connect the female coupling to the male coupling of the 1<sup>st</sup> hose section loaded.
6. Load the remainder of the 1<sup>st</sup> section of hose in the same manner.

**Note: PO's regarding hose loads are for instructional purposes. All hose loads have variations for loading and deploying. The BFST recognizes hose loads and deployment can be accomplished by using different methods and does not intend to evaluate these procedures.**

## **Performance Objective**

Subject: Hose

Task: Deploy a Pre-connect Minuteman Load

Procedure:

1. While facing the hose bed, grasp the nozzle, and pull the entire stack approximately one-third ( $\frac{1}{3}$ ) from the hose bed.
2. Rotate your body facing the direction of travel, allowing the hose stack to rest on your shoulder, with the nozzle at your waist.
3. Walk away from the apparatus, pulling the remainder of the hose demonstrates from the hose bed.

Note: When performed properly, the hose should be evenly distributed on your shoulder.

4. While advancing toward the fire or entry point of structure, deploy the hose by flaking each fold separately, off your shoulder.
5. When hose is fully deployed, signal the driver operator for water and be prepared for movement in the hose line due to charging and pressure.
6. Bleed hose line of any air before continuing and be prepared for a backward force (nozzle reaction) when opening the bale and flowing water.

## **Performance Objective**

**Subject:** Hose

**Task:** Pre-connect attack line; Triple Layer Load (S Load)

**Procedure:**

1. Connect the female coupling to the discharge and extend the hose in a straight line away from the apparatus.
2. Pick up the hose approximately two-thirds ( $\frac{2}{3}$ ) of the distance from the apparatus to the nozzle and carry it back to the apparatus. This will form three layers of hose, stacked one on the other, with a fold at each end.
3. Pick up the entire length of the three layers and begin loading the hose by folding the three layers into the hose bed.
4. Fold the layers at the front and the rear of the bed, laying them on top of the previously laid hose. Make all folds at the rear even with the edge of the hose bed.

**Note:** This can be loaded double stacked (side by side), to accommodate the configuration of the hose bed. Angle the hose and make a fold laying adjacent to the first hose, ensuring all tiers of folds are evenly stacked. Continue loading the hose in the same manner.

5. Continue to lay the hose in the bed in an “S” shape configuration until the entire length is loaded.
6. A fold can be secured to the nozzle with a rope or strap, or by inserting the fold through the bale of the nozzle (The fold must be removed from the bale prior to charging the hose line).

**Note:** PO's regarding hose loads are for instructional purposes. All hose loads have variations for loading and deploying. The BFST recognizes hose loads and deployment can be accomplished by using different methods and does not intend to evaluate these procedures.

## **Performance Objective**

**Subject:** Hose

**Task:** Deploy a Pre-connect Triple Layer Load (S Load)

**Procedure:**

1. While facing in the direction of travel, place the nozzle and fold of the first tier over your shoulder.
2. Walk away from the apparatus, pulling the remainder of the hose demonstrates from the hose bed.
3. Flake the hose fold off your shoulder.
4. Advance toward the fire or entry point of structure.
5. Signal the driver operator for water and be prepared for movement in the hose line due to charging and pressure.
6. Bleed hose line of any air before continuing and be prepared for a backward force (nozzle reaction) when opening the bale and flowing water.

**Note:** The Triple Layer is the quickest to deploy and load. It provides less stress (weight) on the individual, limiting physical exertion and exhaustion in using the load.

## Performance Objective

Subject: Hose

Task: Advance Hose up a Ladder

Note: It is easier, faster, and safer to advance a dry, uncharged hose line up a ladder, than a charged hose line.

### Procedure:

1. Confirm order with command.
2. Flake hose on the ground near the base of the ladder.
3. Facing the ladder, bring the hose up under the arm and across the chest.
4. Lay the hose over the shoulder, resting the nozzle at the small part of your back, on the opposite side work is to be performed.
5. After verifying ladder is safe to climb, ascend the ladder **grasping the rungs** until reaching the desired height.

Note: When advancing hose above the 2<sup>nd</sup> floor, position additional personnel every 10 feet on the ladder. There should be enough hose between personnel, with the hose over the shoulder on the working side, to allow maneuverability without affecting other personnel.

6. Apply a leg lock on the opposite side work is being performed (or attach a harness), as instructed in the Ladder Performance Objective.
7. If available/needed, attach a hose rope tool or a hose strap to secure the hose to the ladder.
8. Bring the nozzle over the shoulder in front of you, ensure the bale is closed, and signal for water to charge the hose line.
9. **Alert personnel below of work to be performed.**
10. Direct the nozzle down and away from the fire, then bleed the hose line and set the desired fog stream pattern if necessary.

Note: Do NOT direct the nozzle across the body to the opposite side work is being performed. This defeats the purpose of having a leg lock on the opposite side.

## Performance Objective

Subject: Hose

Task: Interior Standpipe Connection and Advance Hose up a Stairway

Note: **Detect and verbalize “Gasket, Swivel” for every female coupling throughout the entire process.**

### Procedure:

1. After confirming order with command, bring hose, tools and appliances needed for the operation (a minimum of 100 feet of hose).
2. Connect to the standpipe one (1) floor below the fire floor (If fire is located on the 5<sup>th</sup> floor, connect to standpipe on the 4<sup>th</sup> floor).
3. Remove the cap from the standpipe outlet and detect for any debris or tampering.
4. Open the valve and flush the standpipe to ensure adequate water source, then close the valve.
5. Attach a gated wye to the standpipe outlet. This appliance allows for a 2<sup>nd</sup> attack hose line to be attached. A reducer (verbalize “Gasket” only) can be used but is limited to one (1) attack line.
6. Attach the hose line to the gated wye.
7. The entire hose load should be placed on the shoulder next to the outside wall of the stairway, with the nozzle against your body.
8. Advance up the stairway beyond the fire floor, deploying the hose and laying it against the **outside wall**.

Note: You may pull the hose up, through the center of an open stairway to minimize hose usage.

9. At a desired point, create a curve/loop in the hose and advance down the stairway returning to the fire floor. When descending, deploy the hose to the **inside of the stairway**.
10. When positioned at the door of the fire floor, signal for water to charge the hose line.
11. Check the door for heat with the back of an ungloved hand. Start at the bottom of the door and move your hand toward the top.
12. Bleed the hose line and set the desired fog stream pattern if necessary.
13. Position personnel at every floor and landing (or any point of resistance), to aid in advancing the charged hose line.

## **Performance Objective**

**Subject:** Hose

**Task:** Replace a Burst Section

**Note:** **Detect and verbalize “Gasket, Swivel” for every female coupling throughout the entire process.**

**Procedure:**

1. After confirming order with command, bring two (2) sections of equal diameter hose to replace one (1) section. Both sections should be donut rolls, which provides the easiest and quickest method to replace hose.
2. Apply a hose clamp no closer than five (5) feet from the coupling and on the supply side. Stand on the supply side of the clamp, when applying to the hose.
3. Place the hose centered in the jaws to avoid pinching the hose, then close the clamp to shut off the water supply.

**Note:** **Closing a valve to shut off the water supply is the preferred and safest method.**

4. After the water flow has stopped, break the 1<sup>st</sup> (nearest) coupling as previously instructed. Toss the female coupling of the burst section to the side opposite from the replacement hose rolls.
5. Unwrap the female coupling of the 1<sup>st</sup> hose roll, exposing the male coupling. Connect the female coupling to the male coupling of the clamped hose.
6. Unwrap the female coupling of the 2<sup>nd</sup> hose roll, exposing the male coupling. Connect the female coupling to the male coupling of the 1<sup>st</sup> hose roll.
7. Grasp the male coupling and the connected coupling of the hose rolls, then advance to the 2<sup>nd</sup> coupling of the burst section (this will unroll the replacement hose).
8. Break the 2<sup>nd</sup> coupling as previously instructed. Toss the male coupling of the burst section to the side opposite from the replacement hose.
9. Connect the male coupling to the female coupling to demonstrate the task. Ensure the nozzle bale is closed, then open and remove the hose clamp from the hose.

**Note:** **Testing may involve hose section attached to hydrant (engine) or middle section.**

## **Performance Objective**

**Subject:** Hose

**Task:** Extend a Hose Line with **same** Diameter Hose

**Note:** **Detect and verbalize “Gasket, Swivel” for every female coupling throughout the entire process.**

**Procedure:**

1. After confirming order with command, bring additional sections of equal diameter hose and a hose clamp to the nozzle.
2. Open the nozzle slightly, to allow trickling of water flow.
3. Apply the hose clamp no closer than five (5) feet from the nozzle, as previously instructed.

**Note:** **Closing a valve to shut off the water supply is the preferred and safest method.**

4. After the water flow has stopped, remove the nozzle, attach additional hose line(s), and reattach the nozzle at the end of the additional hose line.
5. Open and remove the hose clamp from the hose.

## **Performance Objective**

**Subject:** Hose

**Task:** Extend a Hose Line with **smaller** Diameter Hose

**Note:** **Detect and verbalize “Gasket, Swivel” for every female coupling throughout the entire process.**

**Procedure:**

1. After confirming order with command, bring additional sections of smaller (1¾ inch) diameter hose, a smaller nozzle, a hose clamp, and a gated wye to the nozzle of the larger diameter hose. The gated wye allows for a 2<sup>nd</sup> hose line to be attached. A reducer (“Gasket” only) can be used but is limited to one (1) attack line.
2. Open the nozzle slightly, to allow trickling of water flow.
3. Apply the hose clamp no closer than five (5) feet from the nozzle, as previously instructed.

**Note:** **Closing a valve to shut off the water supply is the preferred and safest method.**

4. After the water flow has stopped, remove the nozzle on the larger diameter hose. Attach the gated wye or reducer, additional 1¾ inch hose line(s), and the smaller nozzle(s) at the end of the additional hose line(s).

**Note:** **If a breakaway nozzle is used on the larger diameter hose, the nozzle tip shall be removed, and the smaller diameter hose will be attached directly to the breakaway nozzle. The hose clamp and gated wye or reducer, is not needed for this method.**

5. Open and remove the hose clamp from the hose.

## Performance Objective

Subject: Hose

Task: Execute and Operate a Large Handline (2½ or 3-inch) Exposure Loop;  
Keli-Coil

Procedure:

1. After confirming order with command, grasp the nozzle and make a large loop in the hose. The loop should be approximately fifteen (15) feet in diameter. Approximately forty-five (45) feet of the hose line will be utilized.
2. The hose line should cross approximately five (5) feet from the coupling, on the discharge side.
3. Pass the nozzle under the hose approximately two (2) feet, to achieve the appropriate working position.
4. **Directly behind the nozzle, approximately ten (10) feet of the hose line should be straight to aid in controlling the fire stream and reducing nozzle reaction.**
5. Sit on the hose line where it crosses to operate the nozzle.
6. Incidents involving long durations, use a rope or strap to tie a clove hitch where the hose line crosses, to secure the hose in place.

## Performance Objective

Subject: Hose

Task: Supply an FDC (Fire Department Connection)

Note: **Detect and verbalize “Gasket, Swivel” for every female coupling throughout the entire process.**

### Procedure:

1. After confirming order with command, bring hose, tools and appliances needed for the operation.
2. Using 2½ or 3-inch diameter hose, lay the hose with the male coupling toward the FDC and female coupling at the apparatus.
3. Remove the **plugs** on the FDC, then detect for any damage, debris, and clapper valve is present. **Connect the male coupling to the left side of the FDC.**
4. Remove the **caps** on the discharge of the apparatus, **then connect the female coupling to the discharge.** Charge the hose line to supply water to the FDC.
5. **Connect a 2<sup>nd</sup> hose line to the right side of the FDC and to the discharge of the apparatus.** Charge the 2<sup>nd</sup> hose line to adequately supply the FDC.
6. **If the swivels on the FDC are rusted or do not operate freely, there are two (2) options to rectify:**
  - a. Utilize double-male and double-female adaptors to create a functional swivel; or if adaptors are not available,
  - b. Rotate the male coupling of the hose line counterclockwise six (6) turns, then thread the male coupling into the FDC. The hose will lay flat without any twists or kinks.
7. **For a reverse hose lay to the FDC connection, the female coupling will be at the FDC and the male coupling will be at the apparatus and requires the following:**
  - a. **The double-male adaptor must be utilized at the FDC and the double-female adaptor must be utilized at the discharge of the apparatus.**

## Performance Objective

Subject: Hose

Task: Execute and Deploy a Foam Line from the Apparatus

Note: **Detect and verbalize “Gasket, Swivel” for every female coupling throughout the entire process.**

### Procedure:

1. After confirming order with command, bring hose, tools and appliances needed for the operation.
2. Verbalize and select the correct foam concentrate necessary for the type of fire indicated.
  - a. **Class A Fire: Class A Foam**
  - b. **Hydrocarbon Fire: AFFF or FFFP**
  - c. **Polar Solvent Fire: AR/AFFF or FFFP**
3. **Remove the cap on the discharge of the apparatus, then connect a reducer (“Gasket” only). A gated wye is NOT applicable for this evolution.**
4. Attach the in-line eductor to the apparatus or in the hose line. Adjust the eductor metering valve to the same percentage rating indicated for the foam concentrate.
  - a. **Class A: 1%**
  - b. **Hydrocarbon: AFFF 1%, 3% or 6%; FFFP 3%**
  - c. **Polar Solvent: 6%**
5. Attach the proper length of hose, **at least fifty (50) feet and no more than two hundred (200) feet**, to the discharge of the in-line eductor.
6. Attach the nozzle at the end of the hose line.
7. Place the eductor suction hose into the foam concentrate. **The in-line eductor should NOT be more than six (6) feet above the liquid surface of the foam concentrate.**
8. Charge the hose line and open the nozzle fully.
9. **Increase the water supply pressure as required to effect finished foam.**

## Performance Objective

Subject: Hose

Task: Execute and Deploy a Courtyard Lay or Leader Line

Note: **Detect and verbalize “Gasket, Swivel” for every female coupling throughout the entire process.**

### Procedure:

1. After confirming order with command, bring hose, tools and appliances needed for the operation.
2. Remove the cap on the discharge of the apparatus.
3. Connect the required length of 2½ or 3-inch diameter hose to the discharge of the apparatus, **utilizing a minimum of one hundred (100) feet of hose.**
4. Extend the hose to the desired location.
5. Attach a gated wye to the end of the supply hose line, ensuring the valves are closed. This appliance allows for a 2<sup>nd</sup> attack hose line to be attached.
6. Charge the supply hose line.
7. Connect a 1¾ inch attack hose line with a nozzle to the gated wye, utilizing a minimum of fifty (50) feet of hose.
8. Open the gated wye valve to charge the attack hose line.

Note: If known by any other name, inform the field examiners prior to testing.

## **Performance Objective**

**Subject:** Hose

**Task:** Execute and Deploy a portable Master Stream Device; Ground Monitor

**Note:** **Detect and verbalize “Gasket, Swivel” for every female coupling throughout the entire process.**

### **Procedure:**

1. After confirming order with command, bring hose, tools and appliances needed for the operation.
2. Place the portable monitor at the desired location on a solid, level surface.
3. Secure the monitor as necessary.
4. Remove the cap on the discharge of the apparatus.
5. Connect the required length of 2½ or 3-inch diameter hose to the discharge of the apparatus.
6. Extend and connect the hose line(s) to the portable monitor.
7. Position yourself behind the portable monitor to operate properly.
8. Signal for water to charge the hose line.
9. Adjust the nozzle elevation and direction as needed.

## Performance Objective

Subject: Hose

Task: Straight (Storage) Hose Roll

Procedure:

1. After confirming order with command, lay the hose out straight and flat on a clean surface.
2. **Detect and verbalize “Gasket, Swivel”**
3. Roll the male coupling over onto the hose to start the roll.
4. Form a coil that is open enough to allow your fingers to be inserted.
5. Continue to roll the coupling over onto the hose forming an even roll. As the roll increases in size, keep the edges aligned to make a uniform roll.
6. Lay the demonstrated hose roll on its side.
7. With your foot, lightly tamp any protruding coil edges down into the roll.

## Performance Objective

Subject: Hose

Task: Donut Roll

Procedure:

1. After confirming order with command, lay the hose out straight and flat on a clean surface.
2. **Detect and verbalize “Gasket, Swivel”**
3. Grasp the coupling and flip it over creating a bend in the hose. Continue to carry the coupling to the opposite coupling and place it next to each other evenly.
4. Return to the bend in the hose and face the couplings.
5. From the bend, measure approximately 2½ feet on the male side of the hose.
6. Grasp the hose on the male side and start rolling toward the male coupling.
7. Form a coil that is open enough to allow your fingers to be inserted.
8. While rolling the hose, pull the female side of the hose back a short distance to relieve the tension.
9. When the roll approaches the male coupling, lay the roll on its side.
10. To demonstrate the roll, bring the female coupling around the roll covering the male coupling.
11. The female coupling should be positioned directly across (180 degrees) from the male coupling.

## Performance Objective

Subject: Hose

Task: Twin Donut Roll

Procedure:

1. After confirming order with command, lay the hose out straight and flat on a clean surface.
2. **Detect and verbalize “Gasket, Swivel”**
3. Grasp the coupling and turn toward the opposite coupling. **Do NOT flip the coupling over.**
4. Continue to carry the coupling, creating a curve in the hose. This should resemble a curve on a “racetrack.”
5. Place the coupling evenly next to the other.

Note: The couplings can be offset approximately one (1) foot to connect them together when the roll is demonstrated. This will protect the threads on the male coupling.

6. Return to the curve in the hose and face the couplings.
7. Push the curve end over toward the couplings, creating a triangle shape in the hose, and start rolling the hose.

Note: A rope or strap can be inserted in the center of the roll, utilized to secure the shape and for carrying the roll.

8. Continue rolling the hose to the couplings to demonstrate the roll.

## **Performance Objective**

**Subject:** Hose

**Task:** Damaged Hose Roll; Out-of-Service

**Procedure:**

1. After confirming order with command, lay the hose out straight and flat on a clean surface.
2. **Detect and verbalize “Gasket, Swivel”**

**Note:** For consistency of teaching students and not to confuse them during testing, we recommend verifying “Gasket, Swivel” ensuring this is not the issue.

3. Roll the female coupling over onto the hose to start the roll.
4. Form a coil that is open enough to allow your fingers to be inserted.
5. Continue to roll the coupling over onto the hose forming an even roll. As the roll increases in size, keep the edges aligned to make a uniform roll.
6. Lay the demonstrated hose roll on its side.
7. With your foot, lightly tamp any protruding coil edges down into the roll.

**Note:** The following is an alternate method. Tie a rag or mark with tape on the hose (usually where the damage is located on the hose) to easily verbalize it is damaged. Then make a straight roll as previously instructed.

## **Performance Objective**

**Subject:** Hose

**Task:** Supply Hose and Appliances

**Note:** **Detect and verbalize “Gasket, Swivel” for every female coupling throughout the entire process.**

**Procedure:**

1. After confirming order with command, bring hose, tools and appliances needed for the operation.
2. Following directions is paramount for this task.

**Note:** **Students will be asked to supply an Appliance with a male or female coupling, and to supply Hose with an Appliance (Siamese or Wye). Double male and female adaptors may be needed to demonstrate task.**

3. Students must select the correct Appliance and convey the following:
  - a. Components and Functions of Appliance
  - b. Direction of water flow
4. Supply an appliance by attaching the male or female coupling of a hose to the female on the appliance. A double male adaptor shall be utilized when using the female hose coupling.
5. Supply a hose by attaching the male of an appliance to the female hose coupling. In a reverse hose lay, the double female adaptor shall be utilized and attached to the male hose coupling.
6. Students must convey the following regarding all coupling connections:
  - a. Components of the coupling
  - b. Higbee Indicators and Cuts